1. lsblk — View disks and partitions

- -f — displays filesystem types, labels, UUIDs, and mount points.

```
vanam@Uday:~$ lsblk
NAME
    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 388.4M  1 disk
sdb   8:16   0   186M  1 disk
sdc   8:32   0     2G  0 disk [SWAP]
sdd   8:48   0     1T  0 disk /var/lib/docker
                             /mnt/wslg/distro
                             /
vanam@Uday:~$ lsblk -f
NAME
    FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda ext4   1.0
sdb ext4   1.0
sdc swap   1           2d37b9e7-cd35-4e41-ba99-c4076ee822dc                [SWAP]
sdd ext4   1.0         e83198fc-b5c8-4bec-add3-81a6f2aade9b  947.8G     1% /var/lib/docker
                                                                           /mnt/wslg/distro
                                                                           /
```

2. findmnt — Check where filesystems are mounted

- / — shows the mount information for the root filesystem.

```
vanam@Uday:~$ findmnt
TARGET                   SOURCE                  FSTYPE     OPTIONS
/                        /dev/sdd                ext4       rw,relatime,discard,errors=remount-ro,data=ordered
├─/usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
│                        none                    overlay    rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.6.87.2-microsoft-standa
├─/mnt/wsl               none                    tmpfs      rw,relatime
├─/usr/lib/wsl/drivers   drivers                 9p         ro,nosuid,nodev,noatime,aname=drivers;fmask=222;dmask=222,cache=5,access=client,msize=655
├─/mnt/wslg              none                    tmpfs      rw,relatime
│ ├─/mnt/wslg/distro     /dev/sdd                ext4       ro,relatime,discard,errors=remount-ro,data=ordered
│ ├─/mnt/wslg/versions.txt none[/etc/versions.txt] overlay   rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uuid=on
│ ├─/mnt/wslg/doc        none[/usr/share/doc]    overlay    rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uuid=on
│ └─/mnt/wslg/run/user/1000 tmpfs                tmpfs      rw,nosuid,nodev,relatime,size=797204k,nr_inodes=199301,mode=700,uid=1000,gid=1000
├─/usr/lib/wsl/lib       none                    overlay    rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/up
├─/init                  rootfs[/init]           rootfs     ro,size=3981008k,nr_inodes=995252
├─/dev                   none                    devtmpfs   rw,nosuid,relatime,size=3981008k,nr_inodes=995252,mode=755
│ ├─/dev/pts             devpts                  devpts     rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000
│ ├─/dev/shm             none                    tmpfs      rw,nosuid,nodev,noatime
│ ├─/dev/hugepages       hugetlbfs               hugetlbfs  rw,nosuid,nodev,relatime,pagesize=2M
│ └─/dev/mqueue          mqueue                  mqueue     rw,nosuid,nodev,noexec,relatime
├─/sys                   sysfs                   sysfs      rw,nosuid,nodev,noexec,noatime
│ ├─/sys/fs/cgroup       cgroup2                 cgroup2    rw,nosuid,nodev,noexec,relatime,nsdelegate
│ ├─/sys/kernel/debug    debugfs                 debugfs    rw,nosuid,nodev,noexec,relatime
│ ├─/sys/kernel/tracing  tracefs                 tracefs    rw,nosuid,nodev,noexec,relatime
│ ├─/sys/fs/fuse/connections fusectl             fusectl    rw,nosuid,nodev,noexec,relatime
│ └─/sys/kernel/config   configfs                configfs   rw,nosuid,nodev,noexec,relatime
├─/proc                  proc                    proc       rw,nosuid,nodev,noexec,noatime
│ └─/proc/sys/fs/binfmt_misc binfmt_misc         binfmt_misc rw,relatime
├─/run                   none                    tmpfs      rw,nosuid,nodev,mode=755
│ ├─/run/lock            none                    tmpfs      rw,nosuid,nodev,noexec,noatime
│ ├─/run/shm             none                    tmpfs      rw,nosuid,nodev,noatime
│ └─/run/user            none                    tmpfs      rw,nosuid,nodev,noexec,noatime,mode=755
│   └─/run/user          none[/run/user]         tmpfs      rw,relatime
│     └─/run/user/1000   tmpfs                   tmpfs      rw,nosuid,nodev,relatime,size=797204k,nr_inodes=199301,mode=700,uid=1000,gid=1000
├─/tmp/.X11-unix         none[/.X11-unix]        tmpfs      ro,relatime
├─/mnt/c                 C:\                     9p         rw,noatime,aname=drvfs;path=C:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=5,access=client
└─/var/lib/docker        /dev/sdd[/var/lib/docker] ext4     rw,relatime,discard,errors=remount-ro,data=ordered
```

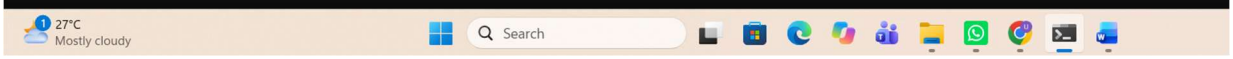
3. blkid — Display filesystem identifiers

```
vanam@Uday:~$ sudo blkid
/dev/sdd: UUID="e83198fc-b5c8-4bec-add3-81a6f2aade9b" BLOCK_SIZE="4096" TYPE="ext4"
/dev/sdb: BLOCK_SIZE="4096" TYPE="ext4"
/dev/sdc: UUID="2d37b9e7-cd35-4e41-ba99-c4076ee822dc" TYPE="swap"
/dev/sda: BLOCK_SIZE="4096" TYPE="ext4"
```

4.lscpu — Inspect CPU details

```
vanam@Uday:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   12
  On-line CPU(s) list:    0-11
Vendor ID:                GenuineIntel
  Model name:             12th Gen Intel(R) Core(TM) i5-1235U
    CPU family:           6
    Model:                154
    Thread(s) per core:   2
    Core(s) per socket:   6
    Socket(s):            1
    Stepping:             4
    BogoMIPS:             4992.00
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss
                          lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq
                           sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnow
                           ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed
                          _ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movd
                           flush_l1d arch_capabilities

Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    288 KiB (6 instances)
  L1i:                    192 KiB (6 instances)
  L2:                     7.5 MiB (6 instances)
  L3:                     12 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-11
Vulnerabilities:
  Gather data sampling:   Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
```




```
vanam@Uday:~$ lscpu | grep '^CPU(s):'
CPU(s):                   12
vanam@Uday:~$
```

5. nproc — Count available processing units

- Default — number of processing units available to the current process.
- `--all` — number of installed processing units reported by the system.

```
vanam@Uday:~$ nproc
12
vanam@Uday:~$ nproc --all
12
vanam@Uday:~$
```

6. lsusb — List USB devices

```
vanam@Uday:~$ lsusb
vanam@Uday:~$
```

7. getconf — Query system configuration values

- `PAGE_SIZE` — memory page size in bytes.
- `LONG_BIT` — size of the `long` integer type in bits.

```
vanam@Uday:~$ getconf PAGE_SIZE
4096
vanam@Uday:~$ getconf LONG_BIT
64
vanam@Uday:~$
```

8. ulimit — Check process resource limits

- `-a` — displays current shell resource limits.
- `-n` — maximum number of open file descriptors.
- `-u` — maximum number of user processes.

```
vanam@Uday:~$ ulimit -a
real-time non-blocking time  (microseconds, -R) unlimited
core file size              (blocks, -c) 0
data seg size               (kbytes, -d) unlimited
scheduling priority                 (-e) 0
file size                   (blocks, -f) unlimited
pending signals                     (-i) 31101
max locked memory           (kbytes, -l) 65536
max memory size             (kbytes, -m) unlimited
open files                          (-n) 10240
pipe size                (512 bytes, -p) 8
POSIX message queues         (bytes, -q) 819200
real-time priority                  (-r) 0
stack size                  (kbytes, -s) 8192
cpu time                   (seconds, -t) unlimited
max user processes                  (-u) 31101
virtual memory              (kbytes, -v) unlimited
file locks                          (-x) unlimited
vanam@Uday:~$ ulimit -n
10240
vanam@Uday:~$ ulimit -u
31101
```

9. ldd — Inspect shared-library dependencies

```
vanam@Uday:~$ ldd /usr/bin/curl
        linux-vdso.so.1 (0x00007fffb9d02000)
        libcurl.so.4 => /lib/x86_64-linux-gnu/libcurl.so.4 (0x00007ba3e0984000)
        libz.so.1 => /lib/x86_64-linux-gnu/libz.so.1 (0x00007ba3e0968000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007ba3e0600000)
        libnghttp2.so.14 => /lib/x86_64-linux-gnu/libnghttp2.so.14 (0x00007ba3e093d000)
        libidn2.so.0 => /lib/x86_64-linux-gnu/libidn2.so.0 (0x00007ba3e091b000)
        librtmp.so.1 => /lib/x86_64-linux-gnu/librtmp.so.1 (0x00007ba3e08fb000)
        libssh.so.4 => /lib/x86_64-linux-gnu/libssh.so.4 (0x00007ba3e0889000)
        libpsl.so.5 => /lib/x86_64-linux-gnu/libpsl.so.5 (0x00007ba3e0875000)
        libssl.so.3 => /lib/x86_64-linux-gnu/libssl.so.3 (0x00007ba3e0556000)
        libcrypto.so.3 => /lib/x86_64-linux-gnu/libcrypto.so.3 (0x00007ba3e0000000)
        libgssapi_krb5.so.2 => /lib/x86_64-linux-gnu/libgssapi_krb5.so.2 (0x00007ba3e0821000)
        libldap.so.2 => /lib/x86_64-linux-gnu/libldap.so.2 (0x00007ba3dffa2000)
        liblber.so.2 => /lib/x86_64-linux-gnu/liblber.so.2 (0x00007ba3e0546000)
        libzstd.so.1 => /lib/x86_64-linux-gnu/libzstd.so.1 (0x00007ba3dfee8000)
        libbrotlidec.so.1 => /lib/x86_64-linux-gnu/libbrotlidec.so.1 (0x00007ba3e0538000)
        /lib64/ld-linux-x86-64.so.2 (0x00007ba3e0a96000)
        libunistring.so.5 => /lib/x86_64-linux-gnu/libunistring.so.5 (0x00007ba3dfd3b000)
        libgnutls.so.30 => /lib/x86_64-linux-gnu/libgnutls.so.30 (0x00007ba3dfb40000)
        libhogweed.so.6 => /lib/x86_64-linux-gnu/libhogweed.so.6 (0x00007ba3dfaf8000)
        libnettle.so.8 => /lib/x86_64-linux-gnu/libnettle.so.8 (0x00007ba3dfaa3000)
        libgmp.so.10 => /lib/x86_64-linux-gnu/libgmp.so.10 (0x00007ba3dfa1f000)
        libkrb5.so.3 => /lib/x86_64-linux-gnu/libkrb5.so.3 (0x00007ba3df956000)
        libk5crypto.so.3 => /lib/x86_64-linux-gnu/libk5crypto.so.3 (0x00007ba3df92a000)
        libcom_err.so.2 => /lib/x86_64-linux-gnu/libcom_err.so.2 (0x00007ba3e0815000)
        libkrb5support.so.0 => /lib/x86_64-linux-gnu/libkrb5support.so.0 (0x00007ba3e052b000)
        libsasl2.so.2 => /lib/x86_64-linux-gnu/libsasl2.so.2 (0x00007ba3df910000)
        libbrotlicommon.so.1 => /lib/x86_64-linux-gnu/libbrotlicommon.so.1 (0x00007ba3df8ed000)
        libp11-kit.so.0 => /lib/x86_64-linux-gnu/libp11-kit.so.0 (0x00007ba3df749000)
        libtasn1.so.6 => /lib/x86_64-linux-gnu/libtasn1.so.6 (0x00007ba3e0515000)
        libkeyutils.so.1 => /lib/x86_64-linux-gnu/libkeyutils.so.1 (0x00007ba3df742000)
        libresolv.so.2 => /lib/x86_64-linux-gnu/libresolv.so.2 (0x00007ba3df730000)
        libffi.so.8 => /lib/x86_64-linux-gnu/libffi.so.8 (0x00007ba3df724000)
```

10. readlink — Resolve symbolic links

```
vanam@Uday:~$ readlink /usr/bin/python3
python3.12
```

Find which Python executable a system command points to.